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Research paper

The predictive role of sedentary behavior and physical activity on adolescent depressive symptoms: A machine learning approach

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ABSTRACT

Objective: This study aims to investigate the predictive value of sedentary behavior and physical activity in adolescent depressive symptoms.
Methods: A total of 2419 adolescent students (grades 7–12) from six administrative regions in China were surveyed. Measures included the Physical Activity Rating Scale for Children (PARS-3), a self-designed questionnaire assessing sedentary behavior among Chinese children and adolescents, and the Children's Depression Inventory (CDI). Machine learning models were trained and tested to predict depressive symptoms based on different types of sedentary behavior, physical activity, and other key variables.
Results: The trained random forest model demonstrated high predictive accuracy (ACC = 90.52 %), with a precision of 92.01 %, recall of 87.95 %, and an F1 score of 0.90. Key predictors of depressive symptoms included sedentary behaviors such as multimedia learning, watching TV, classroom learning, and playing video games. Physical activity also emerged as a significant factor in predicting adolescent depressive symptoms.
Conclusions: The machine learning-based predictive model exhibited strong performance, suggesting that sedentary behavior and physical activity data can effectively predict depression symptoms in Chinese adolescents.

1. Introduction

Depression has been identified by the World Health Organization as one of the most prevalent mental disorders, affecting approximately 3.8 % of the global population, and has emerged as a major public health concern (WHO, 2023). The onset of depression typically occurs during adolescence (Mezuk and Kendler, 2012). Compared to other populations, adolescents are particularly vulnerable to depressive symptoms due to academic pressures and the unique psychological and physiological changes associated with this developmental stage (Choe et al., 2012; Grossberg and Rice, 2023). In China, the prevalence of depressive symptoms in adolescents has increased from 7.4 % in early adolescence to 25 % by the end of adolescence (Fu and Zhang, 2023). Persistent depressive symptoms can have detrimental effects on adolescents' emotions well-being, social interactions, and cognition development (Clayborne et al., 2019; Gregory et al., 2020; Wang et al., 2023). Moreover, they may increase the risk of developing major depressive disorder (Thapar et al., 2012) and suicidal behavior (Leone et al., 2021). Given these potential consequences, early identification of depressive symptoms in adolescents is essential, as it provides critical information for timely and effective prevention and intervention by families and schools. This is particularly important in China, where early detection and targeted prevention strategies for adolescent depression remain a public health priority.

In recent years, increasing academic pressure and the widespread use of mobile electronic devices have led to a significant decline in physical activity among adolescents, accompanied by a marked increase in sedentary behavior. These changes pose substantial risks to both physical and mental health (Pearson et al., 2014). According to the Behavioral Activation Theory, prolonged sedentary behavior reduces engagement in health-promoting activities, such as physical activity, which can have negative consequences for mental well-being and contribute to the onset of depressive symptoms (Collado et al., 2014; Massey-Abernathy and Robinson, 2021). From a neurobiological perspective, physical has been shown to increase levels of key neurotransmitters, such as serotonin and dopamine, thereby alleviating

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depressive symptoms (Krishnan and Nestler, 2008; Lin and Kuo, 2013). Empirical evidence from systematic reviews and meta-analyses suggests that the relationship between sedentary behavior and depressive symptoms may vary depending on the type of sedentary behavior (Wang et al., 2019; Zhang et al., 2022). While most studies have reported a positive association between sedentary behavior and depressive symptoms (Kandola et al., 2020a, 2020b; Opdal et al., 2020), some findings indicate no significant association exists or even a negative correlation (Hallgren et al., 2018; Raudsepp and Vink, 2019). Similarly, systematic reviews and meta-analyses on physical activity suggest that higher levels of physical activity are generally associated with lower depressive symptoms (Rodriguez-Ayllon et al., 2019), however this relationship appears to diminish or even reverse at extremely high levels of physical activity (Kim et al., 2019).

Despite numerous studies on the associations between sedentary behavior, physical activity, and depressive symptoms, the findings remain inconsistent. One potential explanation for these discrepancies is that most previous studies have focused on linear associations, often controlling for extraneous variables or confounding factors, while examining the independent effects of sedentary behavior or physical activity on depressive symptoms (Kidokoro et al., 2021; Pearce et al., 2022; Sampasa-Kanyinga et al., 2020). In addition, sedentary behavior interruption—defined as a the transition from prolonged sitting to standing or walking— has recently gained attention as a key factor in mitigating the adverse effects of prolonged sedentary behavior (Chastin et al., 2015). However, few studies have incorporated this variable into analyses examining the relationships between sedentary behavior, physical activity and depressive symptoms, potentially limiting the predictive power of existing models.

With the rapid advancement of artificial intelligence, machine learning has emerged as a powerful data analysis tool. Compared to traditional linear models, machine learning algorithms can effectively analyze nonlinear relationships among variables (Lee et al., 2018; Su et al., 2023; Zhang et al., 2019), automatically identify key features within datasets, and adapt to changes in data patterns over time (Yang et al., 2020). Furthermore, the precision and accuracy of machine learning models can be enhanced through iterative learning techniques, such as cross-validation and external dataset comparisons, thereby improving predictive performance (Galar et al., 2012).

Recent studies applied machine learning to predict adolescent depressive symptoms using demographic variables, such as gender, age, race, ethnicity, and socioeconomic status (Hawes et al., 2023; Huang et al., 2022). However, to date, no studies have incorporated sedentary behavior and physical activity as predictor variables in machine learning models. Given the well-established associations between these behavioral factors and adolescent depressive symptoms, as well as the relative ease of collecting relevant data in daily life, this study aims to leverage machine techniques to examine the predictive power of sedentary behavior and physical activity in forecasting depressive symptoms among Chinese adolescents in grades 7–12. The findings of this study have important implications. First, they may contribute to the development of a low-cost, convenient, and efficient tool for early depression screening, enabling schools, families, and public health agencies to identify at-risk individuals more effectively. Second, the results may inform school-based mental health programs, family psychological services, and educational policy initiatives, ultimately facilitating the implementation of targeted interventions to prevent adolescent depression.

2. Methods

2.1. Participants

This study distributed electronic questionnaires to students in grades 7–12 from 20 provinces (including municipalities and autonomous regions) across six major administrative regions in China (Northeast, North China, East China, Northwest, Southwest and South Central). The regions surveyed included Shanghai, Fujian Province, Jilin Province, and Tibet Autonomous Region. Classes were selected through convenience and stratified sampling methods, with 600 students per region (100 students per grade). In total 63 schools participated, and 3600 electronic questionnaires were distributed. The returned questionnaires were screened according to the following criteria: 1) incomplete content (e.g., missing answers, incorrect responses, or missing personal information), 2) invalid responses (e.g., multiple answers for single-choice questions, or uniform sedentary time data), and 3) "total sedentary time" exceeding 1 times the interquartile range (IQR) of the outliers. After screening, 2419 valid questionnaires were obtained, comprising 1154 boys and 1265 girls, with a mean age of 15.14 $\pm$ 1.42 years. This study was reviewed and approved by the Committee for the Protection of Human Subjects (Approval Number: HR234–2019). All participants, as well as their parents/guardians, provided written informed consent.

2.2. Measures

2.2.1. Chinese children and adolescents sedentary behavior questionnaire

The study utilized the Chinese Adolescent Sedentary Behavior Questionnaire, developed and revised by Chinese scientists to assess adolescents' daily sedentary behavior over the past week (Li and Liao, 2023; Li et al., 2022). The questionnaire covers various domains, including learning-related sedentary behavior (e.g., classroom learning, doing homework and reading, multimedia learning), screen-based sedentary behavior (e.g., watching TV, playing video games, video activities, social media use), and cultural and recreational sedentary behavior (e.g., drawing, playing musical instruments, playing cards or chess). The questionnaire also assesses interruptions to sedentary behavior, specifically how often adolescents take breaks to move during these activities. The reliability and validity of the questionnaire were satisfactory, with Spearman's correlation coefficients for test-retest the reliability ranging from 0.51 to 0.85 ($p < 0.01$), and intraclass correlation coefficients (ICC) ranged from 0.43 to 0.75. Validity was established by comparing the questionnaire with the Adolescent Sedentary Activity Questionnaire (ASAQ) (Hardy et al., 2007), yielding a Spearman correlation coefficient of 0.786 ($p < 0.01$), confirming acceptable reliability and validity.

2.2.2. Children's depression inventory

The study used the Children's Depression Inventory (CDI), developed by Kovacs (1985) and revised by Wenfeng Wu et al.(Wu et al., 2010). This inventory is designed for children and adolescents aged 7 to 17 years and is widely used in China due to its high reliability and validity, with a Cronbach's alpha coefficient of 0.88. The CDI consists of 27 items, each rated on a three-point scale from 0 to 2, with 0 indicates minimal symptoms and 2 indicates the most severe symptoms. The total score reflects the severity of depressive symptoms, with a threshold score of 19 used to determine the presence of depressive symptoms (Wang et al., 2010).

2.2.3. Physical activity rating scale

The study employed the Physical Activity Rating Scale-3 (PARS-3), developed and revised by Chinese scientist Deqing Liang, to assess adolescents' physical activity over the past week (Liang, 1994). The scale evaluates physical activity across three dimensions: duration, intensity, and frequency, each rated on a five-point scale. The duration dimension is scored from 0 to 4 points, while the intensity and frequency are rated from 1 to 5 points. The physical activity level is calculated as the product of duration, intensity, and frequency, yielding scores ranging from 0 to 100. Higher scores represent higher levels of physical activity. The Cronbach's alpha coefficient for the scale is 0.82.

2.3. Predictive model building

2.3.1. Pre-processing

This study adopted a 5-minute rest standard to statistically analyze interruption in sedentary behavior (Diaz et al., 2016; Healy et al., 2011), which was subsequently used to compute the sedentary time coefficient for each type of sedentary behavior. For example, if sedentary behavior was interrupted every 30 min, the adjusted sedentary behavior time was calculated as: Adjusted Sedentary Behavior Time = Sedentary Behavior Time × 30/35.

Fourteen predictive variables were extracted from the dataset to construct the random forest model. The specific features included are as follows: learning-related sedentary behavior (e.g., classroom learning, doing homework and reading, multimedia learning); screen-based sedentary behavior (e.g., watching TV, playing video games, video activities, social media use); cultural and recreational sedentary behavior; physical activity; age; sex; and sleep duration. Some variables were directly derived from item-level data, (e.g., classroom learning and watching TV), while others were aggregated to form total scores, (e.g., cultural leisure sedentary behavior, learning-related sedentary behavior, and screen-related sedentary behavior). The outcome variable was dichotomized, with '1' indicating the presence of depressive symptoms and '0' indicating the absence of depressive symptoms.

2.3.2. Model selection

The Random Forests (RF) algorithm from the Scikit-learn library in Python was employed for predictive modeling. RF is known for its ability to handle hyperparameters effectively, resist overfitting, and manage class imbalance. By constructing multiple decision trees in parallel, RF enhances the robustness of the model.

As illustrated in Fig. 1, the dataset was partitioned into a training set (70 %) and a test set (30 %). First, multiple decision trees were constructed on the training set, and a Random Forest prediction model was developed through hyperparameter grid search. Second, the model's performance, including accuracy and other metrics, was then evaluated on the test set. To address class imbalance, the Synthetic Minority Oversampling Technique (SMOTE) was applied to generate synthetic data for adolescent samples with depressive symptoms, thus balancing the dataset (Chawla et al., 2002; Han et al., 2005). SMOTE identifies minority class samples and their nearest neighbors to create synthetic data points, mitigating the impact of class imbalance on classification performance. This technique was implemented using the imblearn library in Python 3.10.

2.3.3. Modeling process

Fig. 2 illustrates the decision tree used to distinguish between subjects with and without depressive symptoms. In this figure, blue represents subjects with depressive symptoms (assigned a value of 1), with deeper shades indicating a higher likelihood of depressive symptoms; yellow represents subjects without depressive symptoms (assigned a value of 0), with deeper shades indicating a higher likelihood of not having depressive symptoms. The nodes of the decision tree are divided based on standardized quantitative scales, and node information includes the Gini index (gini), the number of samples reaching the next level (samples), and the number of negative and positive samples in the classification (value).

This study employed cross-validation combined with grid search during the modeling process (Gokten and Uyulan, 2021). To evaluate the model's generalization ability and prevent data leakage, nested cross-validation was used. This method integrates inner cross-validation for hyperparameter tuning and outer cross-validation for model evaluation. In the inner cross-validation, GridSearchCV, a hyperparameter tuning technique, was applied to optimize the random forest model's hyperparameters. The outer cross-validation used stratified K-fold cross-validation (StratifiedKFold), dividing the data into 5 folds, with inner tuning performed on each outer training set. Grid search exhaustively

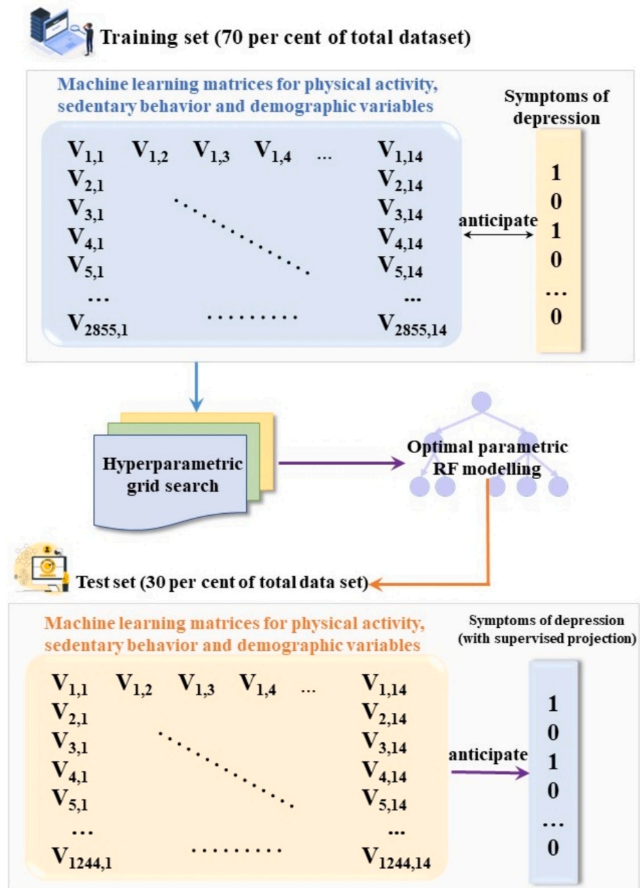

Fig. 1. Overview of model training and procedures.
Notes: In V1,1, V1,2, …, the first number represents the sample size, and the second number represents the number of features.

explored all possible combinations within the given hyperparameter space, returning the best set of hyperparameters, which were used as the optimal parameters for model tuning. The final optimal parameters were: a maximum depth ('max_depth') of 20, minimum samples per leaf ('min_samples_leaf') of 1, minimum samples per split ('min_samples_split') of 2, and the number of trees ('n_estimators') set to 300. Specifically, these parameters indicate that each decision tree has a maximum path length of 20 nodes, each terminal node contains at least 1 sample, non-terminal nodes must split into child nodes with at least 2 samples, and a total of 300 decision trees were constructed for classification and prediction.

2.4. Model evaluation

After the model was established, this study used widely recognized evaluation methods in machine learning to assess the overall classification performance of the model. These methods included true positive (TP), true negative (TN), false positive (FP), and false negative (FN) (Supplemental Table S1).

Based on the above classifications, the following commonly used evaluation metrics were derived: accuracy, precision, recall, F1 score, and area under the curve (AUC). These metrics are defined as follows:

Accuracy is a fundamental metric for evaluating the performance of a classification model. It measures the proportion of correctly predicted instances out of the total number of instances and reflects the overall correctness of the model's predictions. A higher accuracy indicates a better overall classification performance of the model.

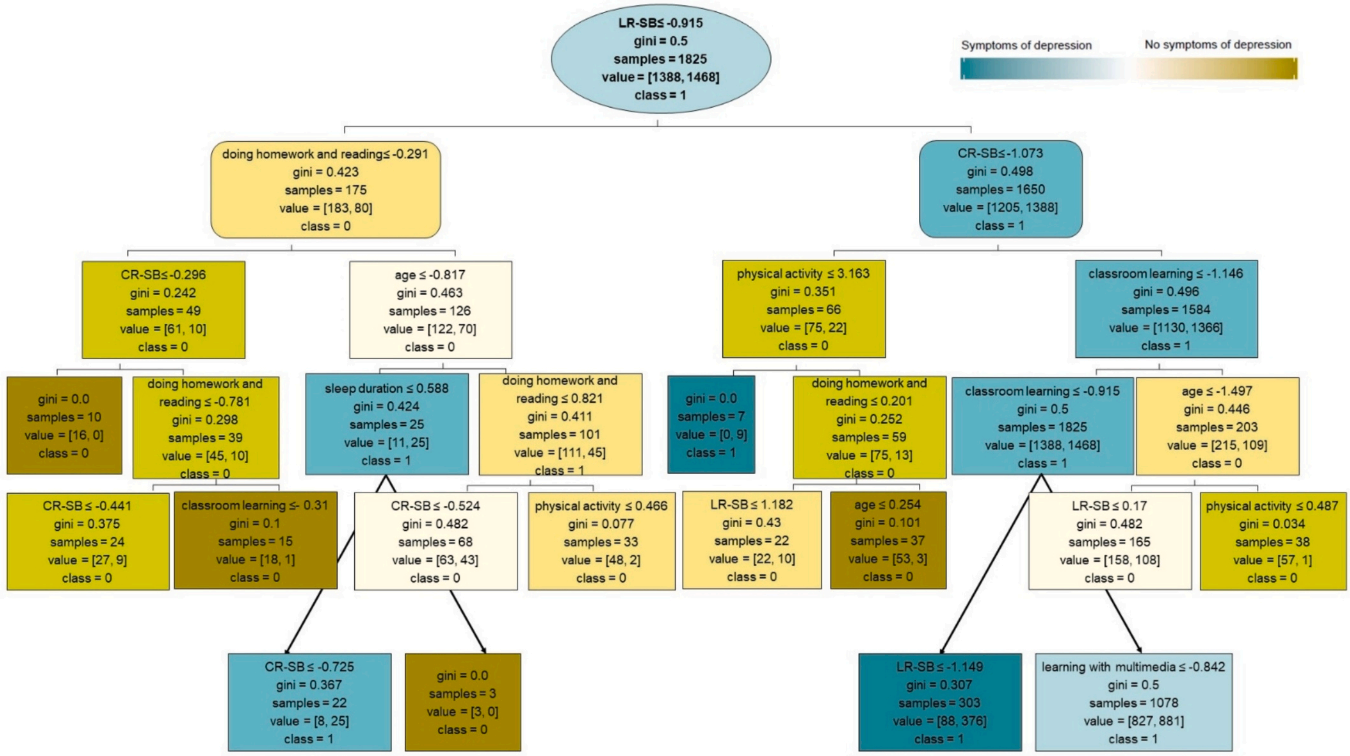

Fig. 2. Decision tree for distinguishing adolescents with and without depressive symptoms.
Abbreviations: gini: Gini coefficient; samples: number of samples reaching the next level; value: number of negative and positive samples in the classification; class 0: adolescents without depressive symptoms; class 1: adolescents with depressive symptoms; LR-SB: learning-related sedentary behavior; CR-SB: cultural and recreational sedentary behavior;

$$accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision measures the proportion of true positive predictions out of all positive predictions made by the model. It is the ratio of correctly predicted positive cases to the total number of positive cases predicted. Higher precision indicates a higher correct detection rate of positive samples by the model.

$$precision = \frac{TP}{TP + FP}$$

Recall measures the proportion of correctly predicted positive cases out of the total number of actual positive cases. Higher recall indicates a higher correct detection rate of positive samples by the model

$$recall = \frac{TP}{TP + FN}$$

The F1 score is the harmonic mean of precision and recall and measures the balance between these two metrics. The closer the F1 score is to 1, the better the model's performance.

$$F1\ score = 2 \times \frac{precision \times recall}{precision + recall}$$

The Receiver Operating Characteristic (ROC) curve and the Area Under the Curve (AUC) are metrics that reflect the characteristics and performance of a classification model. The ROC curve is plotted with the True Positive Rate (TPR, the number of correctly predicted positive instances over the total number of actual positive instances) on the y-axis and the False Positive Rate (FPR, the number of incorrectly predicted positive instances over the total number of actual negative instances) on the x-axis. The AUC value represents the area under the ROC curve. The farther the curve is from the diagonal line (line of equality), the better the model's classification ability. An AUC value closer to 1 indicates higher classification accuracy of the model.

2.5. Post hoc analysis

To systematically evaluate the importance of each feature, this study not only utilized the Mean Decrease in Impurity (MDI) method, which was applied during the random forest modeling process, but also computed SHAP values for each feature as a post-hoc analysis tool. This dual approach addresses the limitations of the MDI method in handling complex feature interactions and nonlinear relationships.

The MDI method assesses the overall impact of each feature on the model by measuring its total contribution to reducing impurity (e.g., Gini impurity) during the decision tree node splits (Kumar et al., 2006). This method provides a straightforward ranking of features within the model framework. However, due to its reliance on internal model statistics, it lacks a nuanced characterization of feature contributions across different samples. To address this limitation, this study further employs SHAP values as post-hoc interpretability tool, aiming to capture the local relative importance of each feature across different samples and to investigate which feature values most significantly influence prediction outcomes.

SHAP values, grounded in cooperative game theory, treat the model's conditional expectation function as the "total surplus." According to the Shapley value principle, a unique and fair contribution is assigned to each feature (Lundberg and Lee, 2017; Shapley, 1953). The process involves calculating the marginal contribution of the target feature across all possible feature orderings, followed by averaging these marginal contributions to derive the SHAP value for that feature.

3. Results

3.1. Sample characteristics

Table 1 shows the differences in demographics, sedentary behavior, and physical activity between adolescents with and without depressive symptoms. Depending on the type of variable, independent samples *t*-tests or chi-squared tests were conducted to assess group differences. The results indicate that adolescents with depressive symptoms exhibited significantly higher total sedentary behavior (T-SB) compared to their nondepressed counterparts. Furthermore, adolescents with depressive symptoms spent significantly more time on screen-based sedentary behavior (SB-SB), particularly engaging in playing video games, video activities, and using social media.

3.2. Random Forest model for predicting adolescent depression symptoms based on sedentary behavior and physical activity

Table 2 summarizes the performance metrics of the Random Forest (RF) model. The model achieved an overall classification accuracy of 90.52 %, with a precision of 92.01 %, a recall of 87.95 %, and an F1 score indicating a well-balanced performance between precision and recall.

Fig. 3 illustrates the receiver operating characteristic (ROC) curve and confusion matrix, where the model attained an area under the ROC curve (AUC) of 0.95, demonstrating excellent discriminative ability between positive and negative cases.

Fig. 4 presents the feature importance ranking based on the mean decrease in impurity (MDI) method. A higher MDI value indicates a greater contribution of the feature to the model's decision-making process. The results suggest that learning-related sedentary behaviors, particularly multimedia learning and classroom learning, play a critical role in predicting adolescent depressive symptoms. Additionally, watching TV and physical activity also exhibit notable influence within the RF model.

Table 1
Sample descriptive statistics ($N = 2419$).

Variable	No ($N = 2040$) [a]	Yes ($N = 379$) [b]	*p*-Value
Age (year)	15.15 ± 1.42	15.09 ± 1.42	0.455
Sex			
Boys	986(48.31 %)	168(44.33 %)	0.154
Girls	1055(51.69 %)	211(55.77 %)	
T-SB (min/d)	526.57 ± 159.28	539.29 ± 172.14	0.006
LR-SB (min/d)	417.86 ± 117.14	422.14 ± 116.43	0.406
Classroom learning	225.00 ± 25.71	225.00 ± 30.00	0.971
Doing homework and reading	162.86 ± 99.85	164.29 ± 110.00	0.496
Learning with multimedia	15.71 ± 27.86	17.14 ± 30.00	0.115
CR-SB (min/d)	31.43 ± 41.43	32.86 ± 47.86	0.304
SB-SB (min/d)	34.28 ± 59.61	37.15 ± 67.13	0.007
Watching TV	8.57 ± 18.57	4.29 ± 17.14	0.120
Playing video games	0.00 ± 14.29	2.86 ± 20.00	0.001
Video activities	8.57 ± 22.86	8.57 ± 25.71	0.026
Social media use	4.29 ± 16.64	5.71 ± 17.14	0.006
Physical activity	6.00 ± 15.00	6.00 ± 13.00	0.479
Depression score	9.00 ± 7.00	24.00 ± 7.00	0.001
Sleep duration			
<8 h	1395(68.38 %)	256(67.55 %)	0.093
8-9 h	508(24.90 %)	86(22.69 %)	
9-10 h	107(5.25 %)	32(8.44 %)	
>10 h	30(1.47 %)	5(1.32 %)	
interruptions to sedentary behavior			0.001
Uninterrupted	98(4.80 %)	38(10.03 %)	
30 min interruption	853(41.82 %)	148(39.05 %)	
45 min interruption	550(26.96 %)	100(26.38 %)	
60 min interruption	539(26.42 %)	93(24.54 %)	

Notes:Age are shown as mean ± SD , other data are shown as median ± IQR (Inter-Quartile Range) or N (%).
Abbreviations: T-SB = total sedentary behavior; LR-SB = learning-related sedentary behavior; CR-SB = cultural and recreational sedentary behavior; SB-SB = screen-based sedentary behavior.
[a] CDI scores < 19.
[b] CDI scores ≥ 19.

Table 2
Model predicts performance of depressive symptoms.

Model	AUC	accuracy	precision	recall	F1 score
Random Forest	0.95	90.52 %	92.01 %	87.95 %	0.90

3.3. Post-hoc analysis and interpretation of the random Forest model

Fig. 5 presents the post-hoc analysis results based on Shapley Additive Explanations (SHAP) values. The top-ranking features identified via SHAP analysis are largely consistent with those derived from the MDI method, with learning-related sedentary behaviors (multimedia learning and classroom learning) emerging as the strongest predictors. Furthermore, screen-based sedentary behaviors, including watching TV and playing video games, also demonstrate high predictive importance. A notable discrepancy between SHAP and MDI analyses lies in the ranking of physical activity features. The SHAP analysis assigns a lower importance to physical activity compared to MDI-based rankings, suggesting potential nonlinear interactions between sedentary behaviors, physical activity, and depressive symptoms.

Fig. 6 shows the Pearson correlation analysis results, revealing significant but relatively weak correlations between multimedia learning, playing video games, and depressive scores In contrast, no significant correlation were observed between classroom learning, watching TV, and depressive scores. Additionally, physical activity did not exhibit a significant correlation with depressive scores. These findings highlight the fundamental differences between linear (Pearson correlation) and nonlinear (Random Forest, SHAP) analytical approaches, suggesting that traditional linear methods may fail to capture complex feature interactions within the dataset.

4. Discussion

This study represents the first attempt to examine the classification performance and feature importance in predicting adolescent depression symptoms using data on sedentary behavior, physical activity, and demographic variables such as sex and age. The machine learning results demonstrate strong performance, with an AUC of 0.95, accuracy of 90.52 %, precision of 92.01 %, recall of 87.95 %, and an F1 score of 0.90. According to previous research, models with an accuracy exceeding 75 % and F1 score >0.80 are considered to perform well (Demsar, 2006). However, it is important to acknowledge that due to the class imbalance in the sample, accuracy may not fully reflect the model's true performance. In this context, AUC is a more robust metric as it evaluates the model's performance all different thresholds, making it more suitable for imbalanced datasets. These findings not only validate the effectiveness of the predictive model but also underscore the potential of sedentary behavior and physical activity data in training accurate models capable of identifying depressive symptoms in adolescents. This has significant implications for the early detection and intervention of adolescent depression, highlighting the potential for targeted prevention strategies.

In this study, we employed two feature importance ranking methods—MDI (Mean Decrease in Impurity) and SHAP (Shapley Additive Explanations)—to assess each the relative contributions of each feature to the depressive symptoms prediction model. Although both methods aim to reveal the roles of features in the model's decision-making process, they differ in their underlying principles. MDI focuses on

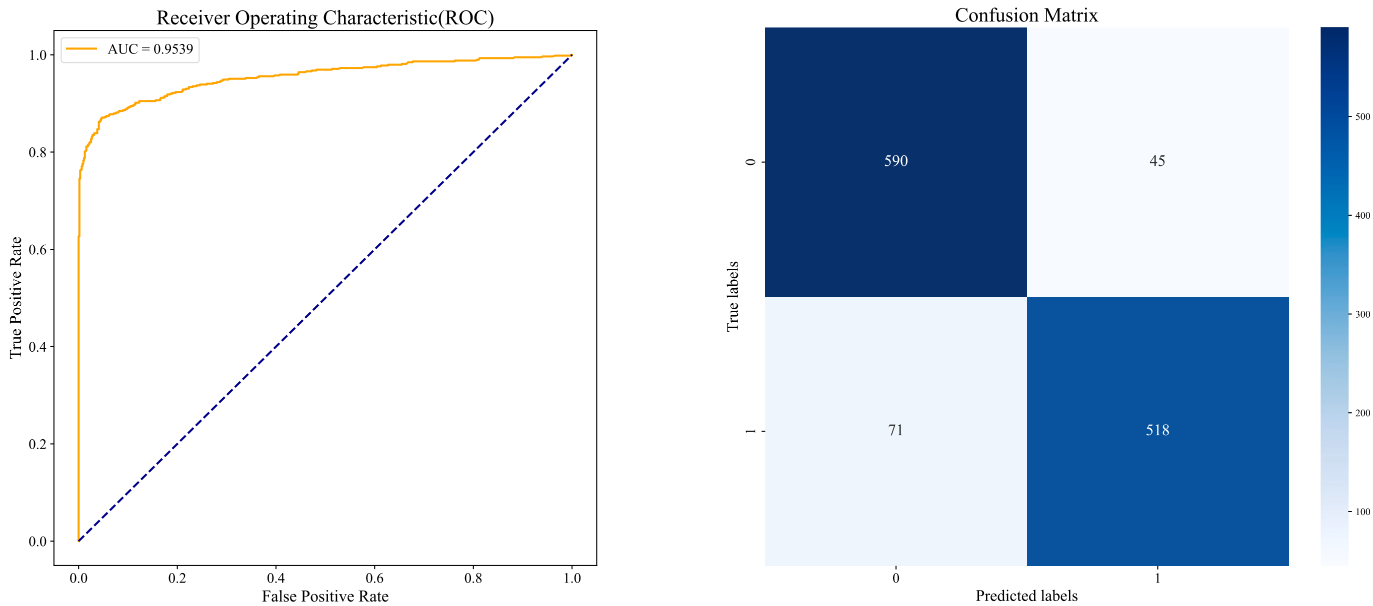

Fig. 3. Visualization of the predictive performance of the Random Forest model for depressive symptoms (ROC plot vs. confusion matrix plot).

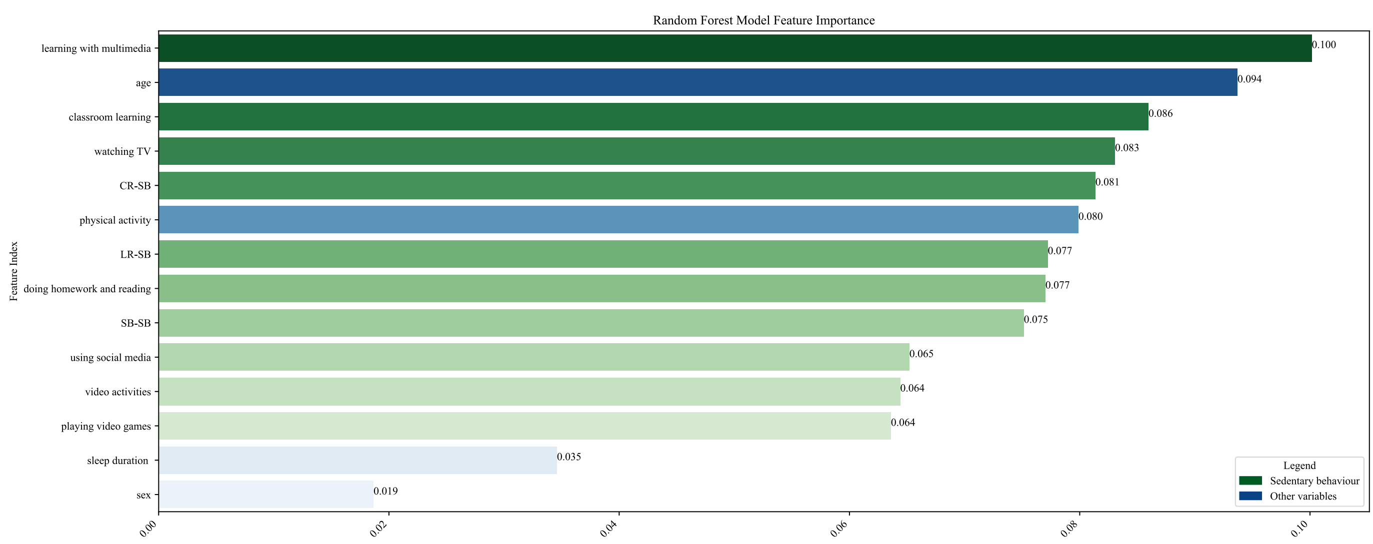

Fig. 4. Feature importance of the random forest model based on MDI approach.
Note: The feature importance ranking (Fig. 4) quantifies each features contribution to reducing model impurity. The x-axis represents MDI values, while the y-axis ranks the features in descending order. Feature categories are colour-coded: blue denotes non-sedentary behavior features, and green represents sedentary behavior features. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

quantifying how features reduce impurity during decision tree training, providing simple and efficient ranking of feature importance. However, MDI may overlook complex feature interactions and nonlinear relationships. In contrast, SHAP values, grounded in a game-theoretic framework, offer a more detailed and precise quantification of each feature's contribution, capturing both linear and nonlinear interactions. This enables more accurate, localized explanations and a deeper understanding of the complex interactions among features in the model.

The feature analysis revealed key predictors of adolescent depressive symptoms. Notably, learning-related sedentary behaviors, specifically multimedia learning and classroom learning, were identified as significant predictors. In China, sedentary behavior is often linked to academic demands, with students spending long hours engaged in learning-related sedentary activities. The pressure of academic performance leads to reduced time for social interaction, increasing the risk of social isolation (Matias et al., 2023). Extended periods of isolation may contribute to the development of depressive symptoms (Zhu et al., 2024). Furthermore, the academic pressure itself can exacerbate sedentary behavior, with prolonged stress contributing to the onset of depression (Lizandra et al., 2016; Zhang et al., 2022).

The study also identified screen-based sedentary behaviors—such as watching TV and playing video games—as important predictors of adolescent depression. This is consistent with previous findings that have linked screen time to depressive symptoms (Maras et al., 2015; Zink et al., 2020). Research has shown that watching TV can disrupt sleep patterns, which in turnincreases the risk of depression (Sun et al., 2022). While playing video games can have short-term relaxing effect, it is often used by adolescents to escape stressful situations, potentially leading to more severe depressive symptoms over time (Cheng et al., 2020; Giardina et al., 2021). Although these behaviors were not found to

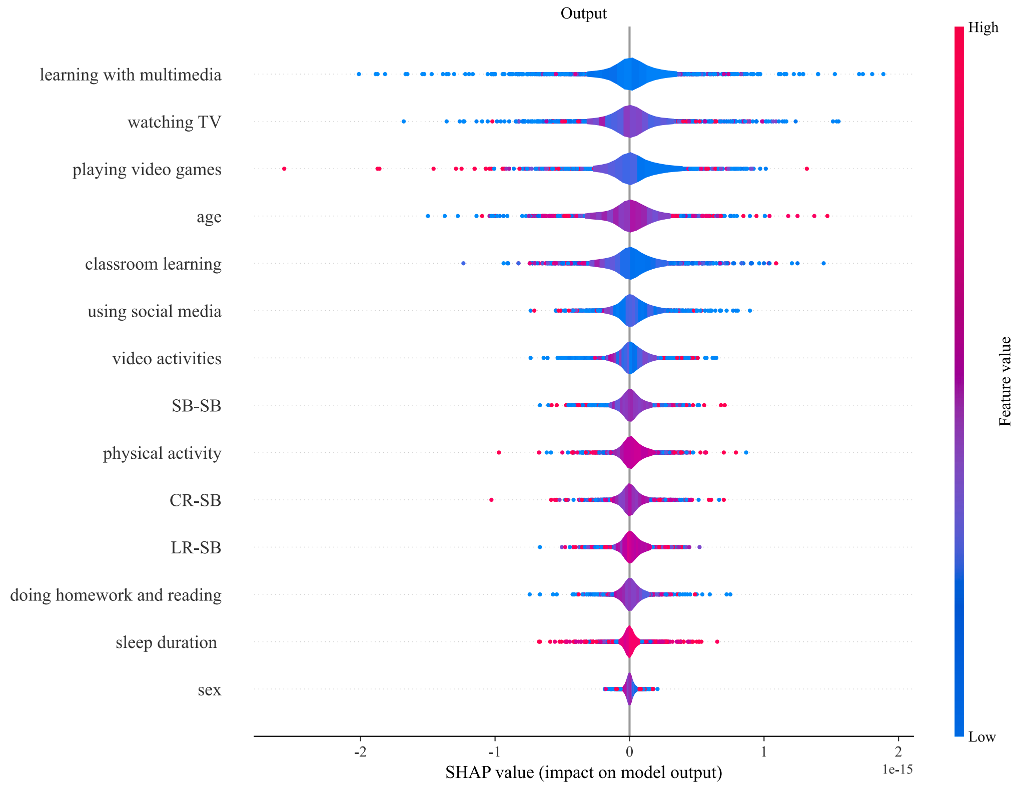

Fig. 5. Post hoc analysis results based on the SHAP method.
Note: Fig. 5 illustrates SHAP value rankings, where each point represents an individual observation. The x-axis denotes the SHAP values, reflecting the magnitude and direction (positive or negative) of a feature's influence on the model's prediction. Point colors range from blue (low feature values) to red (high feature values), providing insights into feature distributions across the dataset. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

be strongly correlated with depression symptoms in a linear fashion, their higher weight in the nonlinear model suggests that the latter may capture more intricate relationships between these behaviors and depressive symptoms, offering insights beyond those provided by traditional linear analysis.

This study also observed that physical activity, though highly ranked in the MDI method, had a lower SHAP value. Despite its modest interaction with other features, physical activity made a significant independent contribution to predicting adolescent depressive symptoms in the RF model, emphasizing its potential as a key intervention variable. Previous studies have highlighted the antidepressant effects of physical activity in adolescents (Jin and Wan, 2022; Recchia et al., 2023). A cohort study found that an additional 60 min of physical activity per day at the ages of 12, 14 and 16 was associated with a significant reduction in depression risk at age 18 (Kandola et al., 2020a, 2020b). Physical activity not only promotes mental well-being but also fosters social connections, mitigating social isolation and further reducing the likelihood of depressive symptoms (Yi et al., 2020).

This study represents a methodological innovation by being the first to apply machine learning approach to the analysis of large-scale data on sedentary behaviors and physical activity among Chinese adolescents. It successfully demonstrates the feasibility of using behavioral data to predict adolescent depressive symptoms, providing a fresh perspective in the field of depression research. Moreover, the high accuracy of the predictive model, which exceeds 90 %, holds practical promise. By incorporating easily accessible information on sedentary behavior and physical activity, the model offers an efficient way to identify and precise predict depressive symptoms. Integrating this model into public health systems could shift adolescent mental health management from reactive to proactive approaches.

However, this study does have some limitations. First, the sample was limited to Chinese adolescents, which may restrict the generalizability of the results. Future studies could consider expanding the sample to other cultural contexts to assess the cross-cultural applicability and robustness of the model. Second, the reliance on self-reported data for sedentary behaviors, physical activity, and sleep may introduce information bias. Future studies could use objective measures, such as accelerometers, to obtain more accurate behavioral data.

5. Conclusion

This study is the first to examine the characteristics of sedentary behaviors and physical activity in adolescents, and to develop a predictive classification model for adolescent depressive symptoms using the random forest algorithm. The model demonstrated outstanding predictive performance, achieving an AUC of 0.95, highlighting its robust predictive power. In-depth feature analysis revealed that sedentary behaviors, such as multimedia learning, watching TV, classroom learning, and playing video games, are identified as important predictors of depressive symptoms in adolescents. Furthermore, physical activity also contributes significantly to the prediction of depressive symptoms. By leveraging the predictive capabilities of machine learning algorithms, this research underscores the potential for schools, families, and society to enhance the identification, prevention, and management of adolescent depression, ultimately contributing to improve adolescent well-being.

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CRediT authorship contribution statement

Lin Li: Writing – review & editing, Writing – original draft,

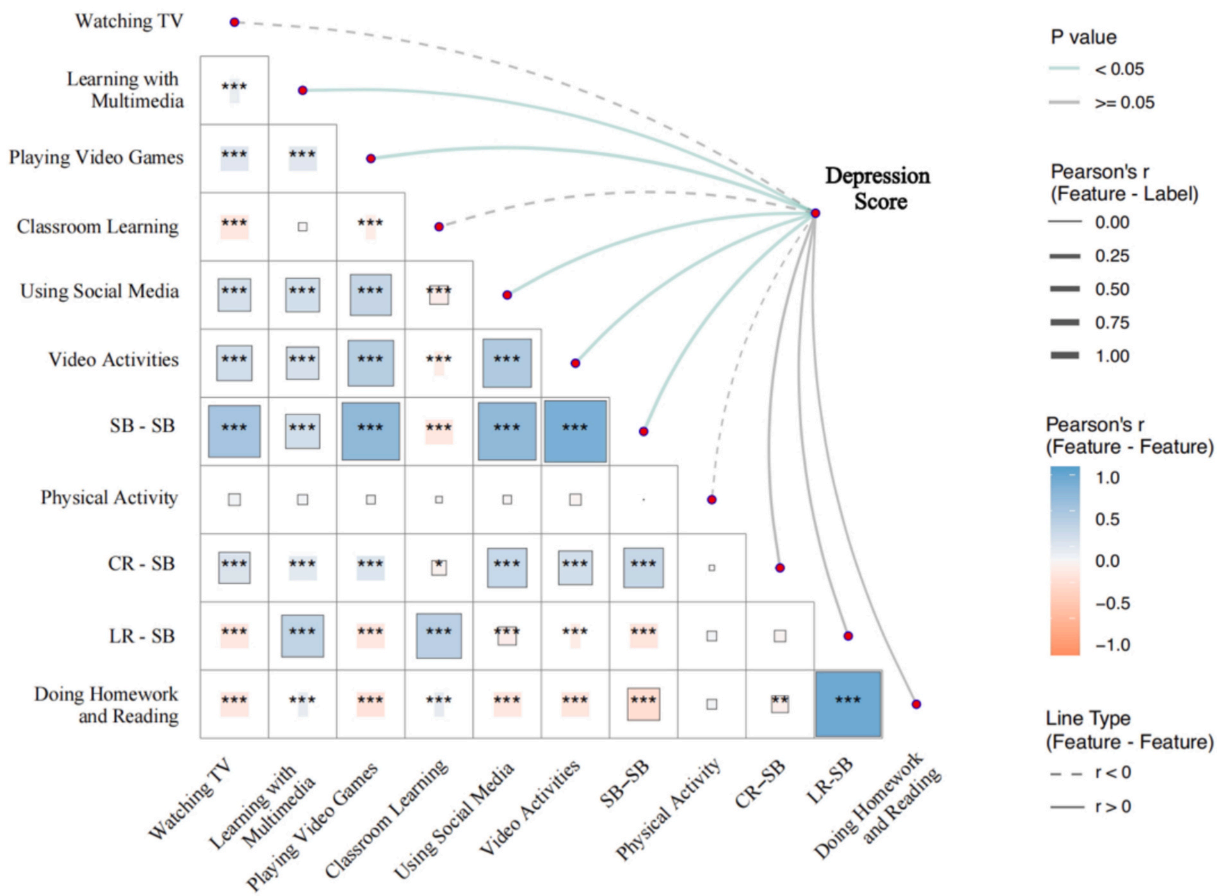

Fig. 6. Results of Pearson's Correlation Analysis.
Note: The heatmap in Fig. 6 visualizes Pearson correlation coefficients. Darker blue shades indicate higher r-value between features. Asterisks (*) denote significance levels: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Line thickness represent the strength of correlation coefficients, while green lines indicate significant correlations ($p <$ 0.05) between sedentary behavior, physical activity, and adolescent depressive symptoms. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

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Declaration of competing interest

The authors declare no conflict of interest.

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Data availability

The datasets generated for this study are available on request to the corresponding author.

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